

C Tasks

Week1:

Q1: C Data Types are defined in three different sections, show them with example of every section with and when to use it?

Q2: 1-show different types of C operators?

2-What is the range of an 8-bit unsigned integer?

3- What is the range of a 16-bit signed integer?

Q3: write a function to take integers (number and the nth bit of number) and return three numbers with this bit of number (cleared, sited, toggled).

Hint (you may use call by reference).

Q4: write code in main to take integer numbers and char to continue the process and it ends when char equal n or N and after this print maximum and minimum taken number and average of this numbers (numbers may be zero or negative or positive) don't use arrays.

Hint (you may use do while statement, you may use floating numbers and print only two digits after decimal point)

Output samples on the next page.

Q5 : 1-Write a C program using switch to display the day of the week based on user input (1 for Monday, 2 for Tuesday, etc.).

2-Write a C program that takes an arithmetic operator (+, -, *, /) and two numbers as input, then performs the operation using switch.

3- Write a switch statement that determines if a given year is a leap year using logical conditions.

A leap year is a year that has 366 days instead of the usual 365 days. The extra day, February 29, is added to keep our calendar aligned with Earth's orbit around the Sun.

✓ Leap Years: 2000, 2004, 2008, 2012, 2016, 2020, 2024

✗ Not Leap Years: 1700, 1800, 1900, 2100

Hint1: (It is divisible by 4 ($\text{year} \% 4 == 0$))

Hint2: (Except for years divisible by 100 ($\text{year} \% 100 == 0$), unless they are also divisible by 400 ($\text{year} \% 400 == 0$)).

Output samples of Q4

```
enter a number : 15
Do you want to continue ? y
enter a number : 20
Do you want to continue ? u
enter a number : 0
Do you want to continue ? t
enter a number : -5
Do you want to continue ? i
enter a number : -5
Do you want to continue ? N

Maximum number: 20
Minimum number: -5
Average of numbers: 5.00

Process returned 0 (0x0)   execution time : 41.188 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter a number : 10
Do you want to continue ? h
enter a number : -3
Do you want to continue ? u
enter a number : 12
Do you want to continue ? N

Maximum number: 12
Minimum number: -3
Average of numbers: 6.33

Process returned 0 (0x0)   execution time : 21.302 s
Press any key to continue.
```

```
Select "C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
enter a number : 5
Do you want to continue ? r
enter a number : -5
Do you want to continue ? n

Maximum number: 5
Minimum number: -5
Average of numbers: 0.00

Process returned 0 (0x0)   execution time : 20.761 s
Press any key to continue.
```